

ASSIGNMENT - CAPSTONE

OVERVIEW

On Friday of last week, local law enforcement executed a search warrant at DataCyte located at 123 Houston Street, San Antonio, TX 78209. The purpose of the search warrant is to seize all electronic devices associated with Christopher Paul Bacon. Mr. Bacon is accused of running a credit card theft operation and selling them on the dark web.

During the seizure, two (2) 64 Gigabyte USB devices were recovered and sent to the lab for analysis. These devices have been imaged and ready for the first round of examination.

Finally, several notebooks were seized and a post-it note was found under Mr. Bacon's keyboard with the phrase "supersecretXXX", where the three "X's" are additional characters that were erased.

INSTRUCTIONS



Case Number: YY-T010

Download all images to your GreymHatter workstation and begin your examination. You will need tools like the Sleuthkit, a hexadecimal reader (e.g. xxd), foremost, strings, etc. If you encounter encryption, you may need to use hashcat and *2john.py tools, obtained at <https://github.com/openwall/john/tree/bleeding-jumbo/run>, veracrypt, etc.

```
# Install foremost
sudo dnf install foremost

# Install veracrypt on Fedora 42-44 (COPR repository)
sudo dnf install dnf-plugins-core fuse fuse-devel -y
sudo dnf copr enable architektapx/veracrypt -y
sudo dnf install veracrypt -y

# Verify veracrypt installed
veracrypt --version

# Check your file system free space
df
```

Mounting a VeraCrypt Volume

If you encounter a VeraCrypt-encrypted volume, you can mount it with:

```
# Create a mount point
sudo mkdir -p /mnt/evidence

# Mount the volume (you will need the password)
sudo veracrypt --text --mount /path/to/image.img /mnt/evidence \
  --password "yourpassword" --pim 0 --keyfiles "" --non-interactive

# Browse the decrypted contents
ls /mnt/evidence
```

```
# When finished, dismount the volume
sudo veracrypt --text --dismount /mnt/evidence
```

Obtaining the offset in the strings command

```
# Hexadecimal offsets
strings -a -t x image.img

# Decimal offsets
strings -a -t d image.img
```

Manage your file system space wisely!

QUESTIONS

Answer the following questions for **each device**.

DEVICE 1 & 2

1. Is this disk partitioned? If so, with what scheme? (MBR or GPT)
2. How many partitions reside on this disk? (Hint: look at the slots section for entries with 000, etc. Not — or Meta)

For each partition (or volume)

1. What is the starting sector and file system associated with the partition?
2. What is the total size (in bytes) for the partition?
3. What is the volume serial number?
4. What is the volume label (name or ID), “None” if no entry?
5. What is the data structure used to track directory or file metadata for this file system? (i.e. FAT Directory Entries, MFT, Inodes)?
6. Does this partition/volume use encryption?
 1. If so, what implementation is used?
 2. If so, what encryption algorithm was used?
 3. If so, what is the password to access the contents?
7. Were there any deleted files on this image?
 1. If so, what is the full path of the file deleted?
 2. If so, based on the extension of the file, what is it?
8. Are there any credit card numbers on this device?
 1. If so, what is the name and full path to the file containing credit card numbers?
 2. If so, what are the contents of this file?
9. Does this volume contain any additional evidence supporting the claims of credit card theft?
10. What do you think this device was used for?

Bonus

1. Is there any hidden data on device 1 that is not visible in any of the mounted partitions? If so, what does it say and where did you find it? Give the exact offset of the data in hexadecimal output.

EVIDENCE FILES

Filename	is4483-capstone-dev1.img
Type	DOS/MBR boot sector; partition 1 : ID=0xee, start-CHS (0x0,0,2), end-CHS (0x3ff,255,63), startsector 1, 134217727 sectors, extended partition table (last)
Size	68719476736 bytes
MD5	DF59681778F78AB2DB0DD8DE8E9BEECC
SHA1	8547E1E43DF12A0550CFC278371618FBBF207C08
SHA256	B65536C155EAD2C2F7976A0432DD6025DA998DF5BBB791F3F7D1C039E7327956

Filename	is4483-capstone-dev2.img
Type	DOS executable (COM, 0x8C-variant)
Size	68719476736 bytes
MD5	FE674640256F452E3493077C064F8E44
SHA1	444C72F2A8335AC787B4F7F148A4B584D1417A0C
SHA256	D5B68BB302F09BBA515AD9E2CD2008FF5DE35FB9781C2A42028CEF26D8309B9C

REQUIRED TOOLS

- All tools used in the course

REQUIRED FILES

- evidence-report-template.docx

DELIVERABLE REQUIREMENTS

- Evidence report with the following sections and requirements
 - Executive Summary
 - Evidence Seized
 - Tools Used
 - Case Information, Observables, Investigator Processes
 - Question answers
 - Conclusion
 - Exhibits (Not required)
 - References (Not required)
 - Photos/screenshots, w/ captions, are required to justify your findings
 - In line with your question answers, not in an appendix/exhibit
 - Must use report template

GRADING RUBRIC

- Evidence report - 100 pts
 - Executive summary - 30 pts
 - Completed answers - 70 pts
 - Answered questions for each item - 1 pts
 - Provided screenshot for each item - 1 pts
 - Used template - zero (0) grade if not used or modified